LABORATORY REPORT 1

Week 4 Laboratory Guide: JavaScript Basics & DOM Manipulation

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Activity: Extending the Week 3 Web Project with JavaScript Interactivity

Introduction

This laboratory activity extends the Week 3 web project by adding JavaScript-based interactivity and DOM manipulation. The activity focuses on a dynamic time display, dark/light theme switching with localStorage, a dynamic To-Do List, keyboard event handling, and form validation using a modal popup. These features make the webpage more interactive while maintaining the semantic structure of the original project.

Objectives

- Display the current time dynamically and update it every second.
- Create a dark/light theme toggle and save the selected theme using localStorage.
- Allow users to add and remove To-Do List tasks dynamically.
- Use a keyboard event to change the page background when a specified key is pressed.
- Validate the contact form email and show validation results inside a modal popup.

III. Features Implemented and Expected Behavior

Feature Implemented	Expected Behavior
Dynamic Header Update	The heading displays the current time and updates every second.
Theme Toggle & Local Storage	The webpage switches between dark and light mode, with the selected theme saved after refresh.
Dynamic Task List	Users can enter a task, add it to the list, and remove individual tasks.
Keyboard Event	Pressing the specified key changes the page background to blue.
Form Validation with Modal	An invalid email shows an error in the modal; a valid email shows “Form Submitted Successfully!”.

Screenshots with Descriptions

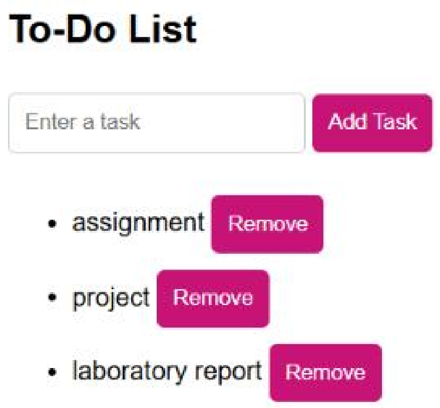

Figure 1 – Dynamic To-Do List

Description: This screenshot shows the Dynamic To-Do List feature. The user can type a task in the input field and click the "Add Task" button. The added tasks, such as "assignment," "project," and "laboratory report," appear in the list. Each task also has its own "Remove" button, allowing the user to delete a task dynamically.

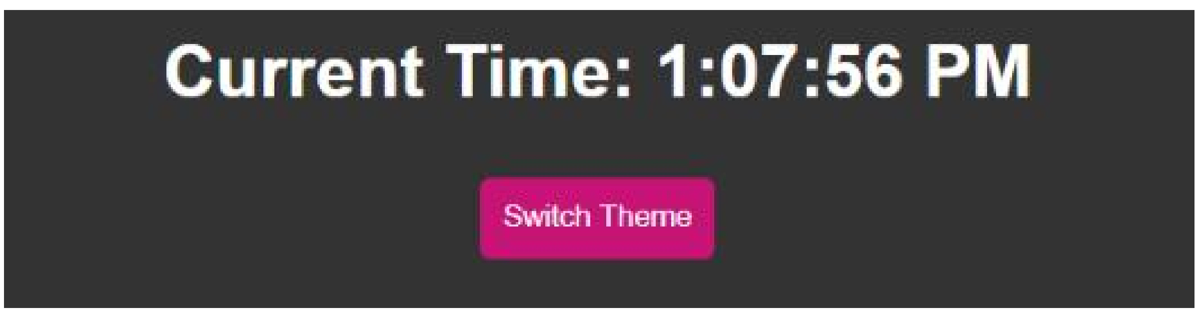

Figure 2 – Dynamic Header and Theme Button

Description: This screenshot demonstrates the Dynamic Header Update. The heading displays the current time, which is automatically updated every second using JavaScript. The "Switch Theme" button is also visible below the time and is used to change the webpage between light and dark modes.

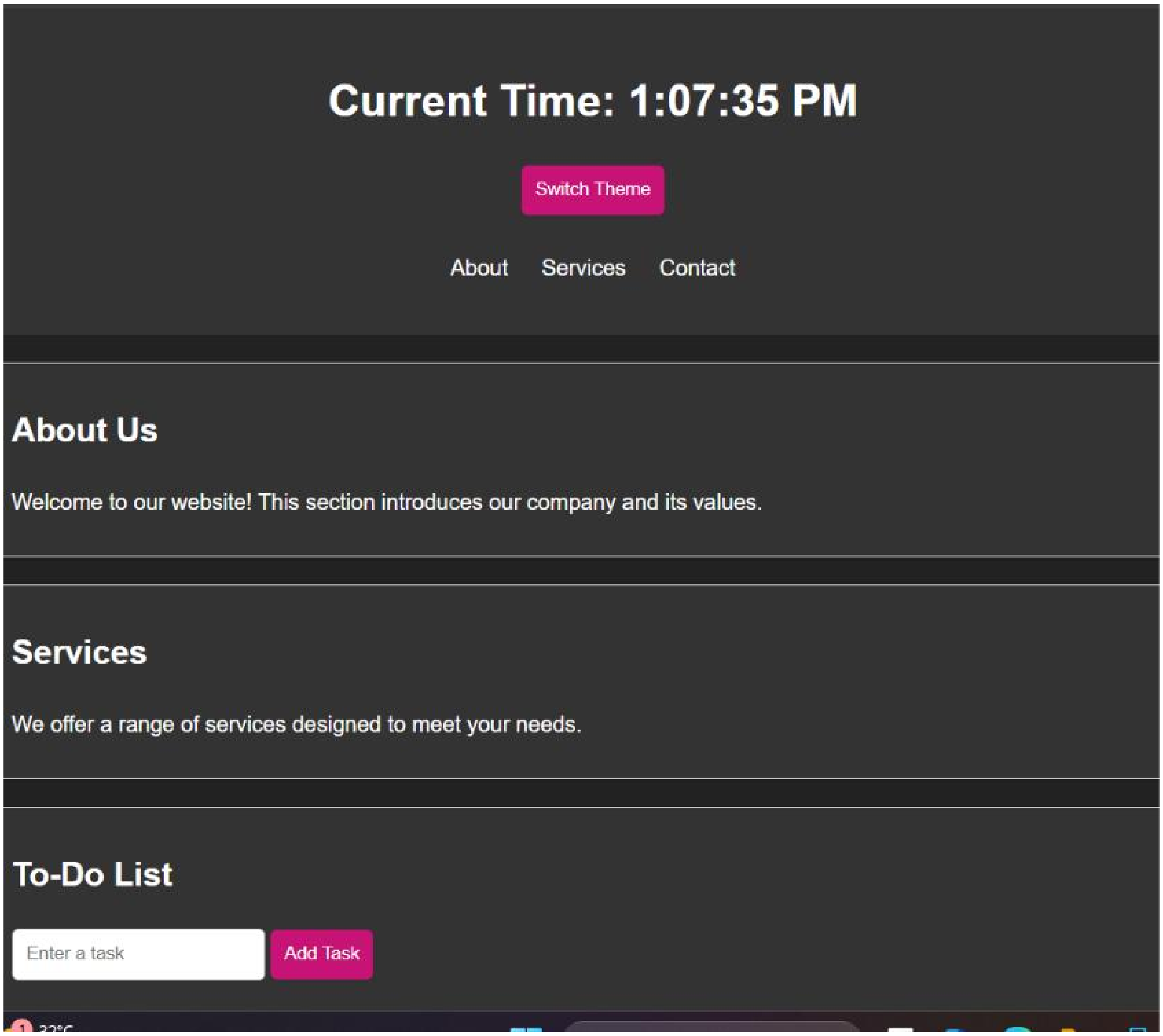

Figure 3 – Dark Mode

Description: This screenshot shows the webpage after the theme has been switched to Dark Mode. The background becomes dark and the text remains visible in a light color. This demonstrates that the theme toggle is working and that the webpage appearance can be changed through JavaScript.

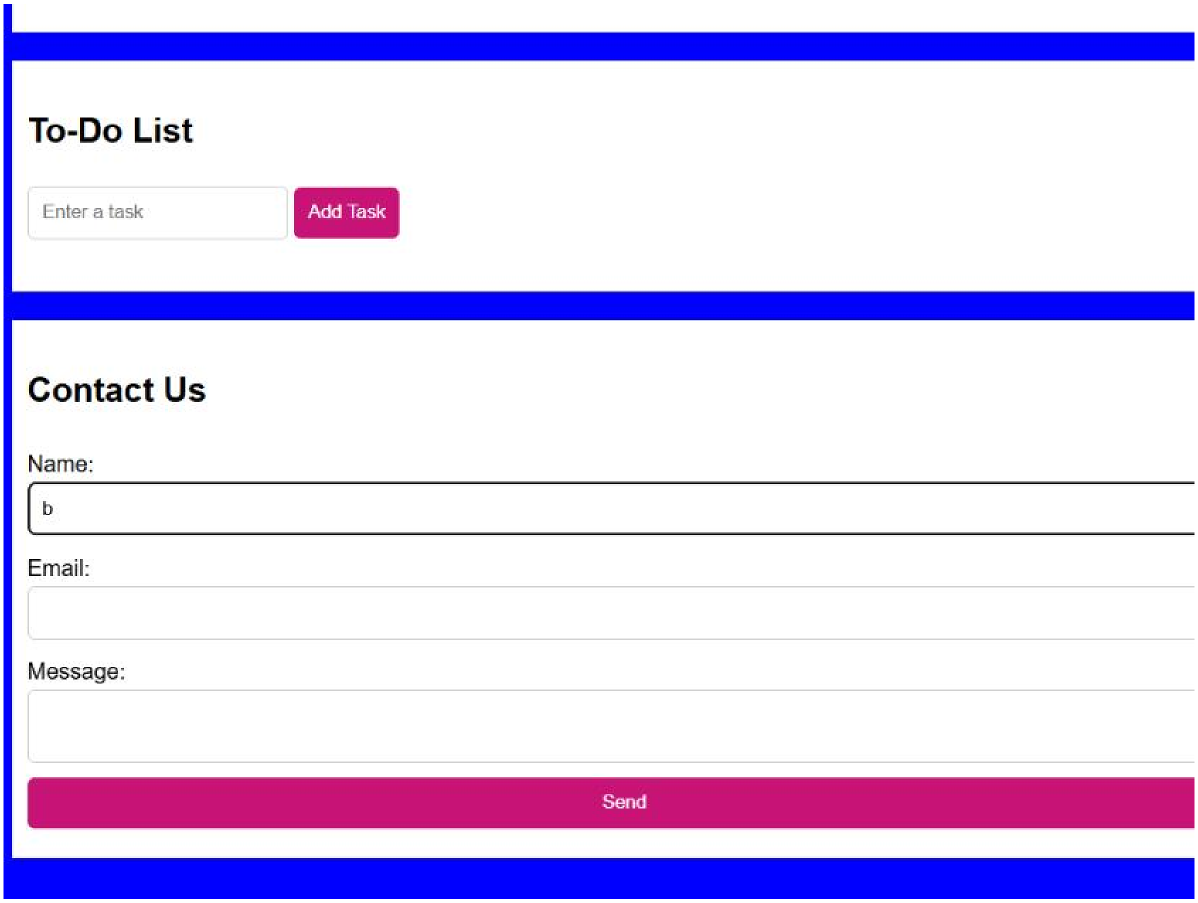

Figure 4 – Keyboard Event

Description: This screenshot demonstrates the Interactive Keyboard Event. When the specified keyboard key is pressed, JavaScript detects the key event and changes the webpage background to blue. This confirms that the keyboard event listener is functioning correctly.

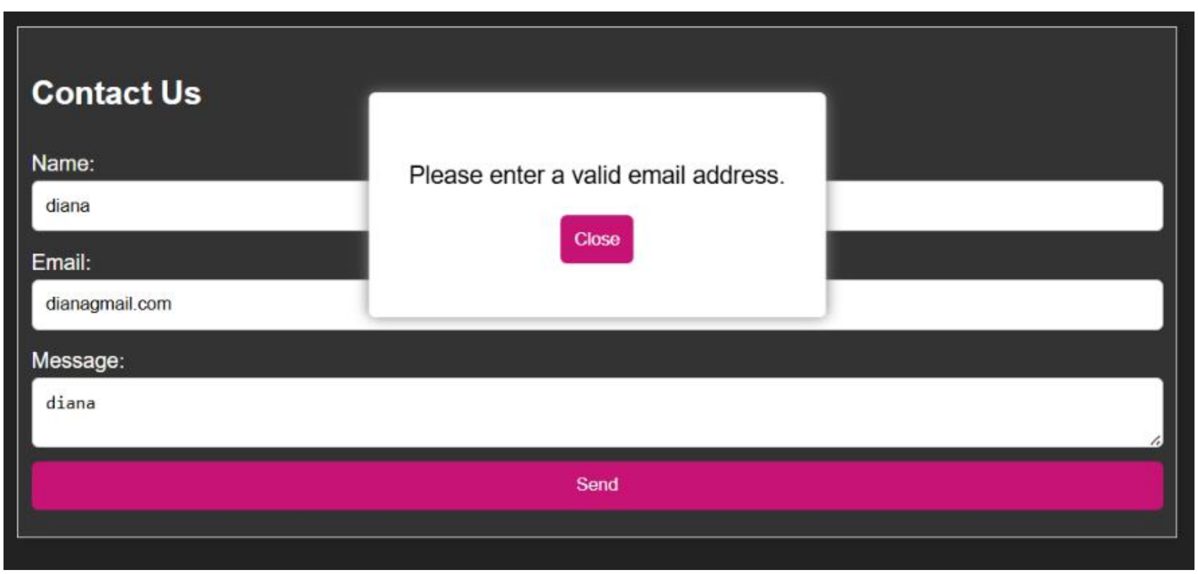

Figure 5 – Invalid Email Validation

Description: This screenshot demonstrates form validation using a modal popup. An invalid email address without the required “@” character was entered. Instead of submitting the form, JavaScript displays the message “Please enter a valid email address.” inside the modal. The user can click the “Close” button to dismiss the message.

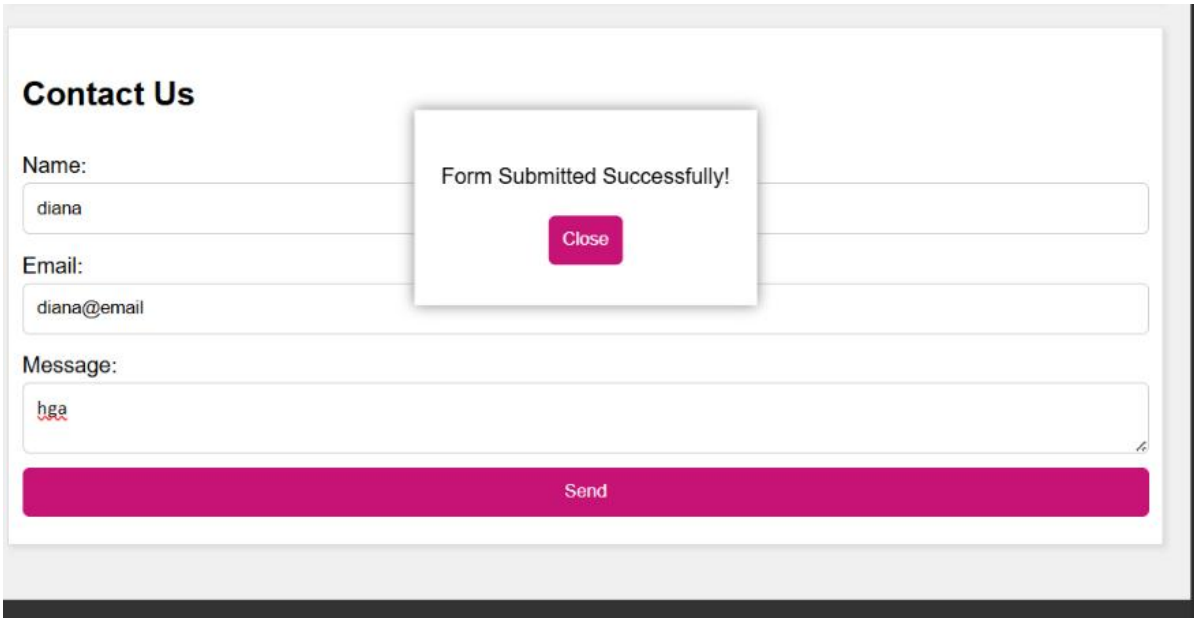

Figure 6 – Successful Form Submission

Description: This screenshot shows the successful form validation process. A valid email address containing “@” was entered, and the form was submitted successfully. JavaScript then displays “Form Submitted Successfully!” inside the modal popup, confirming that the validation condition was satisfied.

JavaScript Concepts Applied

Concept	Application
DOM Manipulation	Accessing and changing HTML elements, creating list items, and updating page content.
Event Listeners	Responding to clicks, keyboard input, and form submission.
setInterval()	Updating the current time every second.
localStorage	Saving the user's selected theme so it persists after refresh.
Conditional Statements	Checking whether the email contains “@” and deciding which modal message to display.
Dynamic Content Creation	Creating To-Do List items and Remove buttons using JavaScript.

Reflection

One of the main challenges I encountered while implementing JavaScript was making all the features work together without affecting the existing Week 3 webpage. At first, it was difficult to understand how JavaScript connects with HTML elements through the DOM. I also had to make sure that the correct element IDs were used so that the buttons, input fields, and form could be controlled properly.

Another challenge was creating the dark and light theme. I learned that changing the page appearance is not only about adding or removing a CSS class. I also needed to use localStorage so that the selected theme would remain even after refreshing the page. This helped me understand how JavaScript can store simple user preferences in the browser.

The To-Do List also helped me practice DOM manipulation. I learned how to get the value from an input field, create new list elements, add buttons, and attach events to those buttons so a task can be removed dynamically. The keyboard event taught me how JavaScript can listen for a key press and respond immediately by changing the background.

For the contact form, I learned how to prevent the normal form submission, check whether the email contains “@”, and display different messages inside a modal. This was more useful than using a simple alert because the modal can be styled and integrated into the webpage.

Overall, this activity helped me understand JavaScript concepts such as event listeners, functions, DOM manipulation, conditional statements, localStorage, keyboard events, form validation, and dynamic content creation. I also learned that testing each feature one at a time makes debugging easier.

Conclusion

The Week 4 laboratory successfully extended the Week 3 webpage with JavaScript interactivity. The implemented features demonstrate practical use of DOM manipulation, event handling, localStorage, keyboard events, timers, dynamic list creation, and form validation. The activity also showed how JavaScript can improve the usability and responsiveness of an existing semantic webpage.